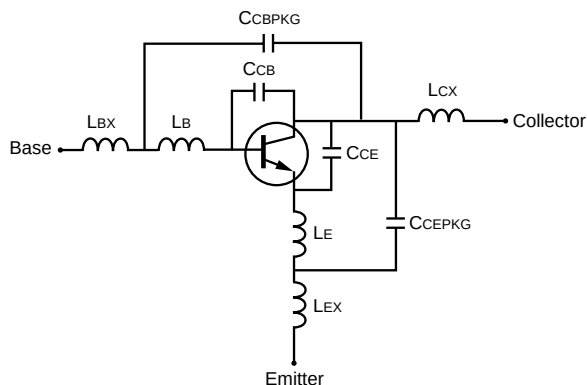


SCHEMATIC



BJT NONLINEAR MODEL PARAMETERS (1)

Parameters	Q1	Parameters	Q1
IS	2.3e-16	MJC	0.28
BF	100.3	XCJC	0.2
NF	0.9917	CJS	0
VAF	31.6	VJS	0.75
IKF	2.2	MJS	0
ISE	3.5e-11	FC	0.7
NE	30	TF	12e-12
BR	13.5	XTF	4
NR	0.9899	VTF	4
VAR	3.9	ITF	0.01
IKR	0.05	PTF	50
ISC	15.4e-12	TR	1.0e-9
NC	30	EG	1.11
RE	1.4	XTB	0
RB	14	XTI	3
RBM	9	KF	0
IRB	0.003	AF	1
RC	1.6		
CJE	1.3e-12		
VJE	0.87		
MJE	0.36		
CJC	0.6e-12		
VJC	0.59		

(1) Gummel-Poon Model

ADDITIONAL PARAMETERS

Parameters	NE58219
CCB	0.03e-12
CCE	0.2e-12
LB	0.75e-9
LE	0.65e-9
CCBPKG	0.17e-12
CCEPKG	0.15e-12
LBX	0.19e-9
LCX	0.7e-9
LEX	0.19e-9

MODEL RANGE

Frequency: 0.05 to 4 GHz
 Bias: $V_{CE} = 1\text{ V to }8\text{ V}$, $I_C = 3\text{ mA to }10\text{ mA}$
 Date: 2/2002

Life Support Applications

These NEC products are not intended for use in life support devices, appliances, or systems where the malfunction of these products can reasonably be expected to result in personal injury. The customers of CEL using or selling these products for use in such applications do so at their own risk and agree to fully indemnify CEL for all damages resulting from such improper use or sale.